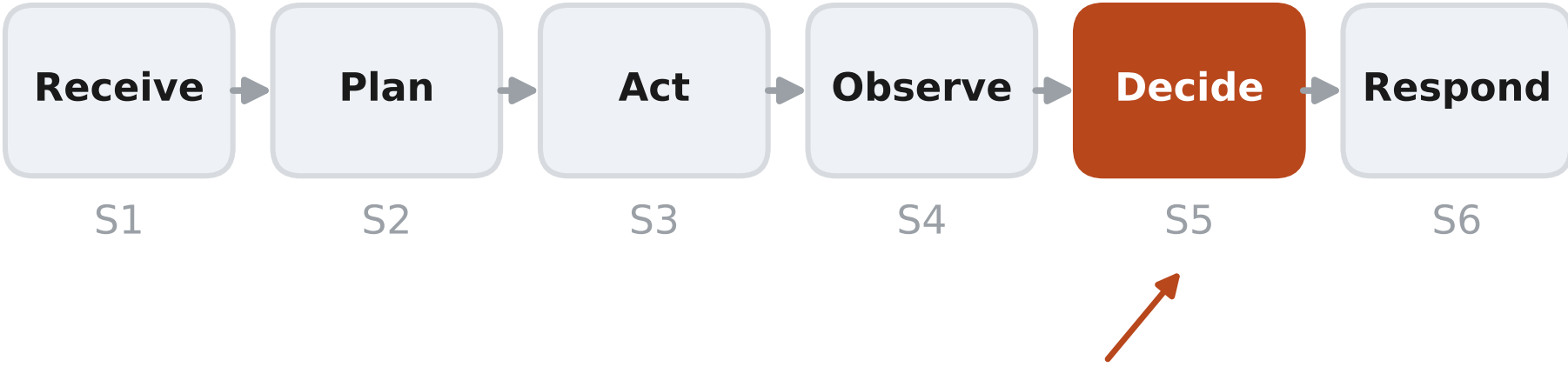observable by callbacks and traces: type, time, inputs, outputs